

Egypt Summer Internship Program 2026

Cloud Build - OCI & Terraform

1. Week 1 - OCI Foundations & Terraform Basics

The first week builds the cloud foundation and introduces Infrastructure as Code. Interns learn OCI core concepts, identity, compartments, compute, storage, and the Terraform workflow.

Oracle University / Learning Content

- Introduction to Cloud Computing
- OCI Global Infrastructure
- Regions, Availability Domains, and Fault Domains
- OCI Core Services Overview
- Compartments
- Identity and Access Management (IAM)
- Compute Services
- Storage Services (Block Volumes – Object storage – File Storage)
- Monitoring and Logging Basics

Terraform / Automation Content

- Introduction to Infrastructure as Code
- Terraform Architecture
- Terraform Installation and Setup
- Terraform CLI Workflow: init, plan, apply, destroy
- OCI Provider Configuration
- Resources
- Variables
- Outputs
- Terraform State File Basics

Hands-on Labs

Lab	Description
Lab 1 - OCI Compute Deployment	Create a VCN, public subnet, and Linux compute instance using OCI Console and attach it to file system and block volume.
Lab 2 - Terraform Basics	Deploy the same basic environment using Terraform.

Assignment

Build Your First OCI Environment

Deliverables

- VCN and public subnet
- Linux compute instance
- Terraform code
- Architecture diagram
- Deployment documentation

KPIs

KPI	Target / Weight
Oracle University Completion	>= 90%
Lab Completion Rate	100%
Assignment Submission	On time
Technical Assessment Score	>= 70%
Attendance	>= 90%

2. Week 2 - OCI Networking & Terraform Intermediate

The second week focuses on OCI networking and secure access patterns. Interns design and automate an enterprise-style network with public and private tiers.

Oracle University / Learning Content

- Virtual Cloud Networks (VCN)
- CIDR planning
- Subnets
- Route Tables
- Security Lists
- Network Security Groups (NSG)
- Internet Gateway & NAT Gateway & Service Gateway
- DNS Services
- Load Balancer
- Bastion service

Terraform / Automation Content

- Data Sources
- Local Values
- Dependencies
- Lifecycle Rules
- Terraform Formatting
- Terraform Validation
- Remote State Concepts

Lab	Description
Lab 1 - Enterprise Network Deployment	Deploy VCN, public subnet, private subnet, internet gateway, NAT gateway, route tables, and security lists and a private instance.
Lab 2 - Secure Access Architecture	Create a bastion session to access the private application server.

Assignment

Enterprise OCI Network Design

Deliverables

- Full OCI network architecture
- Public/private subnet design
- Security list and NSG rules
- Bastion access model
- Terraform code
- Network diagram
- Technical documentation

KPIs

KPI	Target / Weight
Networking Deployment Success	100%
Terraform Code Quality	>= 80%
Documentation Quality	>= 80%
Technical Quiz Score	>= 75%
Architecture Review Score	>= 80%

3. Week 3 - OCI Services, Terraform Automation, OKE.

The third week introduces higher-level OCI services, reusable Terraform modules, modernization practices, and cloud-native deployment using Docker, OCIR, Kubernetes, and OKE.

Oracle University / Learning Content

- Backup and Recovery
- Cost Management
- Docker Fundamentals
- OCI Container Registry (OCIR)
- Kubernetes Fundamentals
- Oracle Kubernetes Engine (OKE)
- OKE Node Pools and Networking

Terraform / Automation Content

- Terraform Modules
- Reusable Infrastructure Design
- Git Integration
- Terraform Workspaces
- Environment Management
- Terraform Best Practices

Lab	Description
Lab 1 - Terraform Modules	Create reusable modules for network, compute, and security.
Lab 2 - Multi-Tier Deployment	Deploy load balancer, two application servers.
Lab 3 - Containerization	Create a Dockerfile, build an image, run it locally, and push it to OCIR.
Lab 4 - OKE Deployment	Deploy OKE cluster, node pool, namespace, NGINX application, Kubernetes service, and OCI load balancer.

Assignment

Containerized Application Deployment

Deliverables

- Assessment of source application dependencies
- Target OCI architecture
- OCI environment deployed using Terraform
- Application files transferred and configured
- Docker image built and pushed to OCIR
- Application deployed on OKE

KPIs

KPI	Target / Weight
Infrastructure Readiness	>= 80%
Validation Success Rate	>= 90%
Terraform Module Quality	>= 80%
OKE Cluster Deployment Success	100%
Docker Image Build Success	100%
Kubernetes Deployment Success	100%
Documentation Quality	>= 80%

4. Week 4 - Final Project

The final week is an end-to-end project that validates the intern's ability to assess, design, build, modernize, validate, document, and present a complete OCI solution.

Final Project Content

Lab	Description
Phase 1 - Assessment	Analyze source environment, identify dependencies.
Phase 2 - Architecture Design	Design the OCI target environment and prepare diagrams.
Phase 3 - Environment Build	Deploy the target environment using Terraform.
Phase 4 - modernization Execution	Containerize the application and complete cutover activities.
Phase 5 - Validation	Perform functional, connectivity, and user acceptance testing.

Assignment

End-to-End OCI Modernization Project

Deliverables

- Terraform repository with modules, variables, outputs, README, and clear structure
- Architecture diagram
- Technical documentation
- Final presentation

KPIs

KPI	Target / Weight
Terraform Code Quality	20%
OCI Environment Deployment	20%
Architecture Design Quality	10%
Microservices deployment	15%
Application Modernization	20%
Documentation Quality	5%
Presentation Skills	5%
Technical Q&A	5%

5. Expected Outcomes

- Design OCI cloud architectures.
- Deploy OCI resources manually and using Terraform.
- Build reusable Terraform modules.
- Implement OCI networking and security best practices.
- Plan and execute cloud modernization projects.
- Create modernization runbooks and validation procedures.
- Containerize applications using Docker.
- Use OCI Container Registry (OCIR).
- Deploy and manage workloads on Oracle Kubernetes Engine (OKE).
- Troubleshoot OCI, Terraform, Docker, Kubernetes, and OKE environments.
- Document and present cloud solutions professionally.